

Bibliometrics and Publications — Felix Dobsław (2026)

1. Bibliometrics

As of April 14th 2026, the PI has published:

- 44 peer-reviewed original articles (12 peer-reviewed journal articles, 32 peer-reviewed conference articles), and 2 other articles.
 - Citations: 694 (Google Scholar, 2026-04-14)
 - h-index: 13 (Google Scholar, 2026-04-14)
 - i10-index: 18 (Google Scholar, 2026-04-14)
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2. Selection of publications (reverse chronological order)

Akbarova, Sabinakhon, and **Dobsław, Felix**, and Gomes de Oliveira Neto, Francisco, and Feldt, Robert. "SETBVE: Quality-Diversity Driven Exploration of Software Boundary Behaviors." *ACM Transactions on Software Engineering and Methodology (TOSEM)*. 2026.

Contribution and Relevance: FD co-supervised and co-authored. SETBVE establishes quality-diversity driven search for systematically exploring software boundary behaviors, advancing the automated extraction of boundary test cases in complex software. The quality-diversity framework offers a scalable, search-based approach to carving out integration points relevant to A2, and its handling of outcome variability in non-deterministic software directly informs the oracle design work in A3.

Dobsław, Felix, and Feldt, Robert, and Yoon, Juyeon, and Yoo, Shin. "Challenges in Testing Large Language Model Based Software: A Faceted Taxonomy." *ACM Transactions on Software Engineering and Methodology (TOSEM)*, special issue on Software Engineering 2030. 2026. (accepted for publication)

Contribution and Relevance: FD coordinated and led the collaboration. This journal version, accepted in TOSEM following the 2025 FSE workshop paper, deepens the faceted taxonomy of challenges in testing LLM-based software and formalizes correctness as a distribution of outcomes rather than a binary property. The DRT project is a natural continuation of this work, particularly regarding semantic non-determinism and the Oracle problem for LLM dependencies in A3.

Akbarova, Sabinakhon, and **Dobsław, Felix**, and Feldt, Robert. "Understanding on the Edge: LLM-generated Boundary Test Explanations." *7th ACM/IEEE International Conference on Automation of Software Test (AST 2026)*. 2026. (accepted for publication)

Contribution and Relevance: FD co-supervised and co-authored. This work investigates how LLM-generated explanations of boundary test behaviour can be used to assess correctness at integration points. It is directly relevant to A3 in DRT, where non-deterministic LLM dependencies require oracle strategies that go beyond binary test verdicts and must articulate the nature of observed variation.

Gren, Lucas, and **Dobsław, Felix**. "The Expert Validation Framework (EVF): Enabling Domain Expert Control in AI Engineering." *CAIN 2026 Industry Track (ICSE Companion '26)*. 2026. (accepted for publication)

Contribution and Relevance: FD co-authored and contributed to the framework design. EVF formalises a mechanism for domain expert control over AI-generated outputs, providing structured correctness assessment in AI engineering. Directly relevant to A3 in DRT, where oracle

strategies for non-deterministic LLM integration points must incorporate interpretable correctness signals beyond binary verdicts.

Axetorn, Jonatan, and Edholm, Felix, and **Dobslaw, Felix**, and Gren, Lucas. “Addressing Trust Requirements in the Design of an Open-Source Multi-Agent LLM-Based Domain-Specific Chatbot.” *Requirements Engineering* (Springer Nature). 2026.

Contribution and Relevance: FD co-authored and contributed to the research design. The paper provides empirical insights into how correctness and trust of LLM components must be specified and validated in a deployed multi-agent system. Relevant to A3 in DRT, where the formalization of non-deterministic dependency tests must account for semantic variability and trust properties of LLM integration points.

Jolak, Rodi, and Karlsson, Simon, and **Dobslaw, Felix**. “An empirical investigation of the impact of architectural smells on software maintainability.” *Journal of Systems and Software* 225 (2025): 112382.

Contribution and Relevance: FD did supervision, co-writing and advisory as the most senior researcher. Highlights challenges of maintainability and particularly testability in often poorly structured projects, strengthening the need for DRT that focuses on isolatable carved-out units as dependency tests.

Dobslaw, Felix, and Feldt, Robert. “Automated Boundary Identification for Machine Learning Classifiers.” *Proceedings of the 17th ACM/IEEE International Workshop on Search-Based and Fuzz Testing*. 2024.

Contribution and Relevance: FD led the work, and did most of the writing, and all of the experiments. Automatically identifying boundary behaviour for ML software includes correctness variance and confidence due to statistical non-determinism — a mandatory perspective for a generic formalization, as the non-determinism differs substantially from LLM produced.

Dobslaw, Felix, and Feldt, Robert, and Gomes de Oliveira Neto, Francisco. “Automated black-box boundary value detection.” *PeerJ Computer Science* 9 (2023): e1625.

Contribution and Relevance: FD led the work, and did most of the writing and all of the experiments. Heuristic search to formalize/generalize and implement the first-ever process of automatically identifying boundaries for traditional software. Relevant as it required defining a complex formerly unmapped process, just like for DRT, with overlap in A1, A2, and A4.

Dobslaw, Felix, Ruiyuan Wan, and Hao, Yuechan. “Generic and industrial scale many-criteria regression test selection.” *Journal of Systems and Software* 205 (2023): 111802.

Contribution and Relevance: FD did all the work — received help in terms of access to industrial data. Formalized the RT problem into a general framework using multi-objective search and extraction of optimal bounds by integrating convex solving. Investigating it empirically with tailored search methods is relevant for the DRT project throughout all activities.

Holmberg, Mats, and **Dobslaw, Felix**. “An industrial case study on GUI testing with RPA.” *2022 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2022.

Contribution and Relevance: FD supervised and guided student and co-authored. Relevant for the project as the industrial work with end-to-end tests revealed the many integration points that are implicitly tested in RT, but that are as of today not easily isolated when failures occur. DRT would lead to shorter fault-finding times in the industry.

3. Publications 2018–2026

3.1 Original Articles

Akbarova, Sabinakhon, and **Dobslaw, Felix**, and Gomes de Oliveira Neto, Francisco, and Feldt, Robert. “SETBVE: Quality-Diversity Driven Exploration of Software Boundary Behaviors.” *ACM Transactions on Software Engineering and Methodology (TOSEM)*. 2026.

Dobslaw, Felix, and Feldt, Robert, and Yoon, Juyeon, and Yoo, Shin. “Challenges in Testing Large Language Model Based Software: A Faceted Taxonomy.” *ACM Transactions on Software Engineering and Methodology (TOSEM)*, special issue on Software Engineering 2030. 2026. (accepted for publication)

Axetorn, Jonatan, and Edholm, Felix, and **Dobslaw, Felix**, and Gren, Lucas. “Addressing Trust Requirements in the Design of an Open-Source Multi-Agent LLM-Based Domain-Specific Chatbot.” *Requirements Engineering* (Springer Nature). 2026.

Muhammad, Laiq, and **Dobslaw, Felix**. “Automatic techniques for issue report classification: A systematic mapping study.” *Automated Software Engineering* (Springer). 2026. (accepted for publication)

Jolak, Rodi, and Karlsson, Simon, and **Dobslaw, Felix**. “An empirical investigation of the impact of architectural smells on software maintainability.” *Journal of Systems and Software* 225 (2025): 112382.

Thilderkvist, Eva, and **Dobslaw, Felix**. “On current limitations of online eye-tracking to study the visual processing of source code.” *Information and Software Technology* 174 (2024): 107502.

Dobslaw, Felix, and Feldt, Robert, and Gomes de Oliveira Neto, Francisco. “Automated black-box boundary value detection.” *PeerJ Computer Science* 9 (2023): e1625.

Dobslaw, Felix, Ruiyuan Wan, and Hao, Yuechan. “Generic and industrial scale many-criteria regression test selection.” *Journal of Systems and Software* 205 (2023): 111802.

3.2 Conference Articles

Gren, Lucas, and **Dobslaw, Felix**. “The Expert Validation Framework (EVF): Enabling Domain Expert Control in AI Engineering.” *CAIN 2026 Industry Track (ICSE Companion '26)*. 2026. (accepted for publication)

Akbarova, Sabinakhon, and **Dobslaw, Felix**, and Feldt, Robert. “Understanding on the Edge: LLM-generated Boundary Test Explanations.” *7th ACM/IEEE International Conference on Automation of Software Test (AST 2026)*. 2026. (accepted for publication)

Dobslaw, Felix, and Feldt, Robert, and Yoo, Shin, and Yoon, Juyeon. “Challenges in Testing Large Language Model Based Software: A Faceted Taxonomy.” *2025 Proceedings of Foundations of Software Engineering (FSE), 2030 Software Engineering*. ACM, 2025.

Dobslaw, Felix, and Sundberg, Leif. “CodeNoCode Predictive Modeling: Co-Development with LLMs for Transparency and Inclusivity.” *1st Workshop on Human-Centered AI for SE (FSE Companion '25)*. ACM, 2025.

Larsen, Knud Ronau, and Edvall, Magnus, and Truong Ho-Quang, and **Dobslaw, Felix**, and Jolak, Rodi. “An LLM Assistant for Software Project Onboarding.” *International Workshop on Envisioning the AI-Augmented Software Development Life Cycle (FSE Companion '25)*. ACM, 2025.

Velander, Johanna, and Otero, Nuno, and **Dobslaw, Felix**, and Milrad, Marcelo. "Eliciting and Empowering Teachers' AI Literacy: The Devil is in the Detail." *International Conference in Methodologies and Intelligent Systems for Technology Enhanced Learning (MIS4TEL)*. Cham: Springer Nature Switzerland, 2024.

Gomes de Oliveira Neto, Francisco, and **Dobslaw, Felix**. "Building Collaborative Learning: Exploring Social Annotation in Introductory Programming." *Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training (ICSE SEET '24)*. ACM/IEEE, 2024.

Dobslaw, Felix, and Feldt, Robert. "Automated Boundary Identification for Machine Learning Classifiers." *Proceedings of the 17th ACM/IEEE International Workshop on Search-Based and Fuzz Testing (SBFT)*. 2024.

Rico, Sergio, and **Dobslaw, Felix**, and Öberg, Lena-Maria. "Starting collaborations between SMEs and researchers in software engineering." *International Conference on Software Business (ICSOB)*. Cham: Springer Nature Switzerland, 2023.

Haapasaari Lindgren, Marcus, and Persson, Jon, and Jolak, Rodi, and **Dobslaw, Felix**. "COMET: A ML-Based Tool for Evaluating the Effectiveness of Software Design Communication." *2023 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*. IEEE, 2023.

Dobslaw, Felix, and Bergh, Peter. "Experiences with remote examination formats in light of GPT-4." *Proceedings of the 5th European Conference on Software Engineering Education (ECSEE)*. ACM, 2023.

Dobslaw, Felix, and Angelin, Kristian, and Öberg, Lena-Maria, and Ahmad, Awais. "The gap between higher education and the software industry — A case study on technology differences." *Proceedings of the 5th European Conference on Software Engineering Education (ECSEE)*. ACM, 2023.

Dobslaw, Felix, and Feldt, Robert. "Similarities of Testing Programmed and Learnt Software." *2023 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2023.

Cvitic, Petya Hristova, and **Dobslaw, Felix**, and Gomes de Oliveira Neto, Francisco. "Investigating software testing and maintenance of open-source distributed ledger." *2023 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. IEEE, 2023.

Holmberg, Mats, and **Dobslaw, Felix**. "An industrial case-study on GUI testing with RPA." *2022 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2022.

Hokka, Heidi, and **Dobslaw, Felix**, and Bengtsson, Jonathan. "Linking developer experience to coding style in open-source repositories." *2021 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. IEEE, 2021.

Åhs, Fredrik, and Mozeliuss, Peter, and **Dobslaw, Felix**. "Artificial intelligence supported cognitive behavioral therapy for treatment of speech anxiety in virtual reality environments." *ECIAIR 2020*. Vol. 2. Academic Conferences and Publishing International Limited, 2020.

Gomes de Oliveira Neto, Francisco, and **Dobslaw, Felix**, and Feldt, Robert. "Using mutation testing to measure behavioural test diversity." *2020 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2020.

Dobslaw, Felix, and Vallin, Morgan, and Sundström, Robin. "Free the Bugs: Disclosing Blocking Violations in Reactive Programming." *2020 IEEE 20th International Working Conference on*

Source Code Analysis and Manipulation (SCAM). IEEE, 2020.

Dobslaw, Felix, and Gomes de Oliveira Neto, Francisco, and Feldt, Robert. "Boundary value exploration for software analysis." *2020 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2020.

Dobslaw, Felix, and Feldt, Robert, and Michaëlsson, David, and Haar, Patrik, and Gomes de Oliveira Neto, Francisco, and Torkar, Richard. "Estimating return on investment for GUI test automation frameworks." *2019 IEEE 30th International Symposium on Software Reliability Engineering (ISSRE)*. IEEE, 2019.

Feldt, Robert, and **Dobslaw, Felix**. "Towards automated boundary value testing with program derivatives and search." *Search-Based Software Engineering: 11th International Symposium, SSBSE 2019*. Springer International Publishing, 2019.